NeKoRAG Project Overview

NeKoRAG is a hybrid retrieval-augmented generation system. It combines dense vector search using ChromaDB with bge-m3 embeddings, and sparse keyword search using BM25 with jieba tokenization. The two retrieval paths are merged via Reciprocal Rank Fusion (RRF), which requires no weight tuning and is robust to different score distributions. After fusion, a cross-encoder reranker (BGE-reranker-v2-m3) re-scores the candidates for final ranking. Context expansion follows linked-list pointers between adjacent chunks to retrieve surrounding context, preventing information loss across chunk boundaries. The system supports streaming SSE generation via DeepSeek API, with automatic source citation in responses.

Key Features:

- Hybrid retrieval (Dense + BM25 + RRF)
- Cross-encoder reranking for precision
- Context expansion via linked-list traversal
- Streaming SSE generation with citations
- PDF / Markdown / TXT document support
- Docker containerization with health checks